

On the Relationship between Inter-Area Modes and Power System Inertia

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Abstract—Increased use of wind and solar power is leading to reduction of inertia in several power systems. Lower inertia (or kinetic energy) has a significant effect on various dynamic phenomena in power systems. One of the affected phenomena is electromechanical inter-area oscillations. The characteristics of inter-area modes change as synchronous generators participating the modes are replaced by converter connected wind and solar farms. This paper analyses the characteristics (frequency and damping) of the main inter-area mode in the Nordic power system (i.e., the 0.3–0.4 Hz mode between Finland and Sweden) over a period of 5 years. The estimated frequency and damping of the mode are then compared with the system inertia from the same period. The results show that the frequency of the inter-area mode has a strong correlation with system inertia. The damping of the mode has practically no correlation with inertia. Understanding these relationships is important and may be used in the future in evaluating system stability. Furthermore, the relationship between inter-area mode frequency and inertia may be utilized in estimating the total kinetic energy of the system in the future.

Keywords—Inertia, inter-area oscillation, power system monitoring.

I. INTRODUCTION

Large scale integration of wind and solar power is leading to decreasing inertia (or kinetic energy) in various power systems. These resources are replacing traditional sources of inertia, i.e., synchronous generators, and they do not usually provide any additional inertia in the system. For example in the Nordic power system, very low inertia situations have been observed recently [1] and the inertia levels are expected to decrease even more substantially in the coming decades [2].

Decreasing inertia changes the dynamic behavior of the power system. Different dynamic phenomena are affected, one of them being inter-area electromechanical oscillations. In some power systems, like the Nordic system, inter-area oscillations limit the transmission capacity of certain transmission corridors and might risk the system stability in certain situations. Thus, it is important to understand the characteristics of inter-area modes now and in the future.

This paper analyses, using measurements, how the characteristics of the main inter-area mode between Finland and Sweden (0.3–0.4 Hz mode) has changed during 2017–2021 (5 years). A mode meter [3] is used to analyze the characteristics (frequency and damping of the mode) from ambient PMU measurements. When the frequency and damping of the mode have been estimated during the 5-year analysis period, the estimates are compared with the total kinetic energy of the system. The results show that the natural frequency of the inter area mode is strongly correlated with

the inertia of the system (negative correlation). As expected, the frequency is lower during winter times (high inertia) and higher during summertime (low inertia), but a clear relationship between the values can be observed in hourly estimates during the whole analysis period. The damping of the mode, however, is not correlated with inertia. Furthermore, the natural frequency of the mode has a slightly rising trend during the years and system inertia has a decreasing trend.

Understanding the relationship between inter-area modes and inertia is highly important since inertia is expected to decrease substantially in the future, thus having a significant effect on the characteristics of the inter-area modes, and consequently, system stability. Also, when the characteristics of inter-area modes change, this might require retuning of controllers such as power system stabilizers. Furthermore, understanding the relationship between inter-area modes and inertia might enable the use of ambient modal identification methods (mode meters) for real-time evaluation of system inertia.

This paper is structured as follows. Section II provides background information about the Nordic power system and presents the analysis methods – Bayesian modal identification method and inertia evaluation system. Section III presents the analysis results and provides discussion on the main findings. Section IV concludes the paper.

II. BACKGROUND

A. Nordic Power System and the Studied Inter-Area Mode

The Nordic power system consists of synchronized power systems of Finland, Sweden, Norway, and eastern Denmark. The southern part of the system contains the majority of the load and the largest generators, including nuclear plants in Finland and Sweden. There is significant hydro power production in the northern part of the system, and wind power production is increasingly being incorporated into the network, especially in the north. The kinetic energy of the system is typically in the range of 140 GWs and 260 GWs, however, recently as low as 110 GWs values have been observed [1].

In this paper, the Finland-Sweden inter area mode (0.3–0.4 Hz) is studied. In this mode, the generators in the southern regions of Finland and those in southern Sweden and Norway, oscillate against each other. Thus, the mode includes a considerable portion of the Nordic system's generators and large amount of inertia contribute to this mode. [4] Fig. 1. presents the Nordic power system and the Finland-Sweden mode.

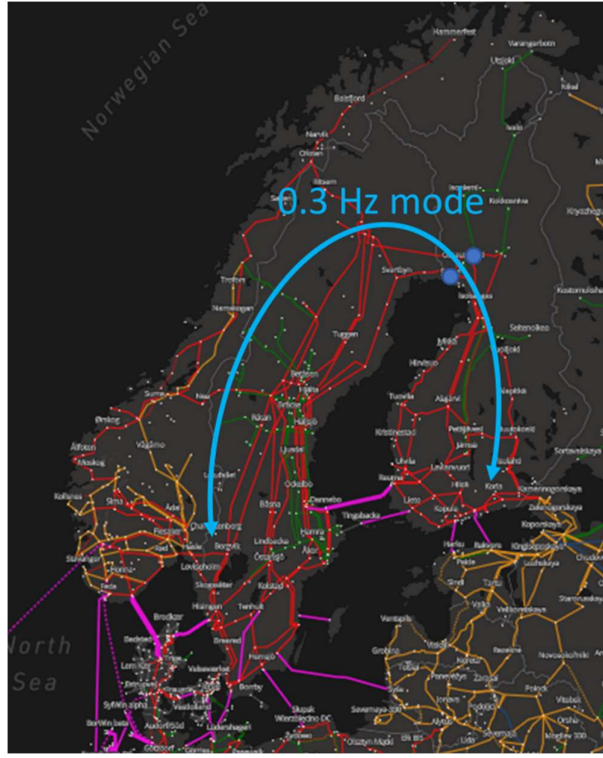


Fig. 1. The Nordic power system and the studied inter-area mode path (the blue arrow curve) [9] and the measurement locations for the 400 kV Finland-Sweden AC interconnectors (dark blue dots, discussed in Section II B).

B. Mode Meter Analysis

There are several methods in the literature for identifying the characteristics of electromechanical modes from ambient measurements [3], [5]–[8]. Ambient measurement based mode meters identify the characteristics of electromechanical modes continuously (online/real-time).

In this paper, the Bayesian method [3] has been used. The method first calculates a Fast Fourier Transform of the input data and identifies the modal parameters based on Bayes Theorem in terms of a probability distribution conditional on the data and modeling assumptions. The flow chart of the methods is presented in Fig. 2. The details of the Bayesian method, including the theoretical background, equations and parameter selection are presented in [3], [5].

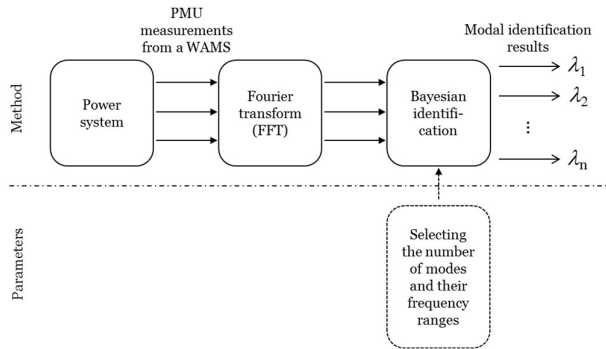


Fig. 2. A flow chart of the Bayesian modal identification method when applied to ambient power system measurement data [3], [5].

The active power flows of two northern 400 kV AC interconnectors between Finland and Sweden (Fig. 1 presents the measurement locations) were used as an input to the method. Fig. 3 presents a sample excerpt of the ambient measurement data. The data were recorded at a 10 Hz sample rate. One hour of data segments were analyzed one at a time from every hour between 2017–2021 (5 years). The natural frequency and damping ratio of the Finland-Sweden inter-area mode were identified with the Bayesian method.

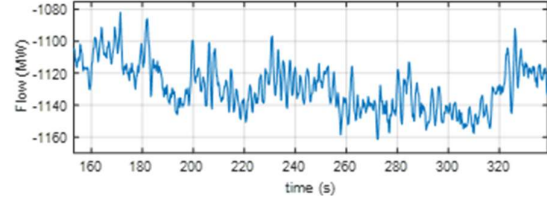


Fig. 3. Example of ambient PMU measurement data (active power flow from Finland to Sweden via the two northern 400 kV AC interconnectors) used as input in the Bayesian modal identification method.

C. Real-Time Kinetic Energy Estimation System

The Nordic Transmission System Operators (TSOs) have implemented a SCADA/EMS-based system for estimating the kinetic energy of the Nordic power system. This system provides online assessments of the system's kinetic energy by summing up the kinetic energies of generators using their known values and generator circuit breaker status information. The system includes generators at all voltage levels but does not take into account contributions from motors, which is expected to be rather small. There are also some small generators that are not measured and therefore are not included in the estimates. [10]

III. RESULTS

A. Estimation Results

Fig. 4 shows the estimated hourly natural frequency of the Finland-Sweden inter-area mode, the damping ratio of the mode and the kinetic energy of the Nordic system. The modal parameters were estimated with the Bayesian method described in Section II B and the kinetic energy was estimated with the approach presented in Section II C.

As shown by the result in Fig. 4, the natural frequency of the inter area mode between Finland and Sweden has a yearly cycle. The frequency is highest during summer periods and lowest during winter periods. There is also a slight increasing trend in the frequency.

The kinetic energy also has a yearly cycle. It is highest in winter periods and lowest during the low load situations of the summer periods (however, this might change in the future). The kinetic energy has a slight decreasing trend during the analysis period.

The damping ratio has been on average very high (mean value is 18 %) during the analysis period. The damping does not have a clear yearly cycle.

B. Relationship between the Modal Parameters and Kinetic Energy

Fig. 5a presents the kinetic energy of the Nordic system and the natural frequency of the Finland-Sweden inter-area mode in the same figure. To illustrate the curves in the same figure, the means of the series have been removed and their

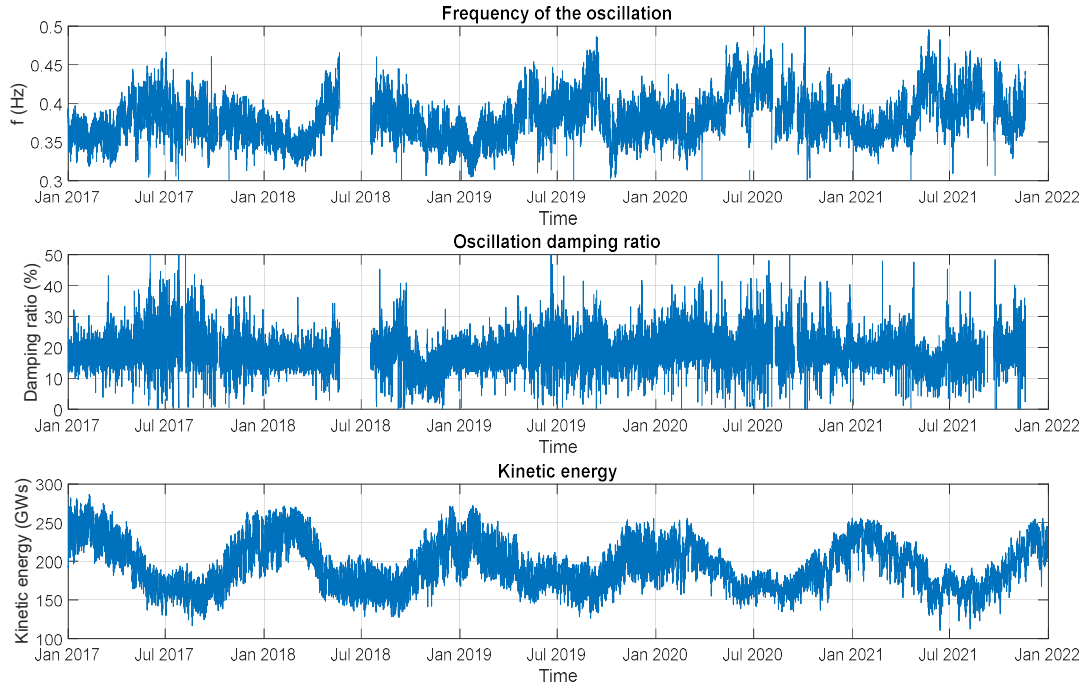


Fig. 4. The estimates of the Finland-Sweden inter-area mode natural frequency, damping ratio of the mode and the kinetic energy of the Nordic power system. The empty segments in the frequency and damping ratio estimates are due to missing measurement data.

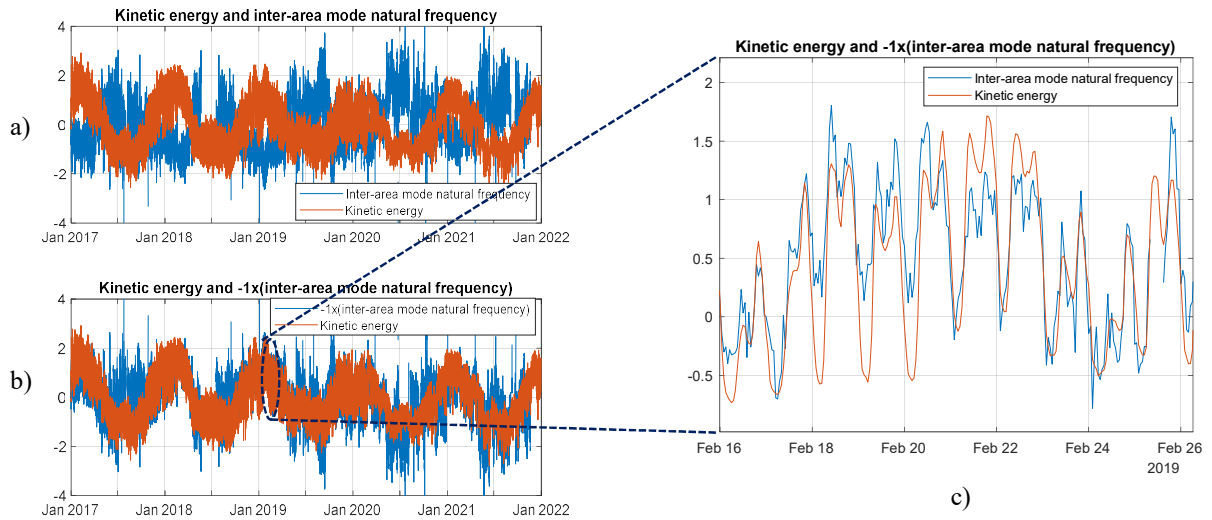


Fig. 5. The kinetic energy and natural frequency of the Finland-Sweden mode (means removed and standard deviations scaled to 1), b) same curves, but natural frequency has been multiplied with -1 to illustrate the relationship of the values, c) zoomed plot from Fig. 5b.

standard deviations have been scaled to 1. Fig. 5b presents the same curves as Fig. 5a, but now the natural frequency has been multiplied with -1 in order to present the relationship between the curves more clearly. Fig. 5c presents a zoomed segment from Fig. 5b.

Fig. 6 shows the same kinetic energy estimates and the natural frequency estimates as in Fig. 5a as a scatter plot. It also presents a linear fit between the two variables.

As shown by Figs. 5 and 6 there is a clear relationship between the natural frequency of the inter-area mode between Finland and Sweden and the kinetic energy of the Nordic

power system. In fact, the correlation coefficient between the two variables was -73 % (negative correlation).

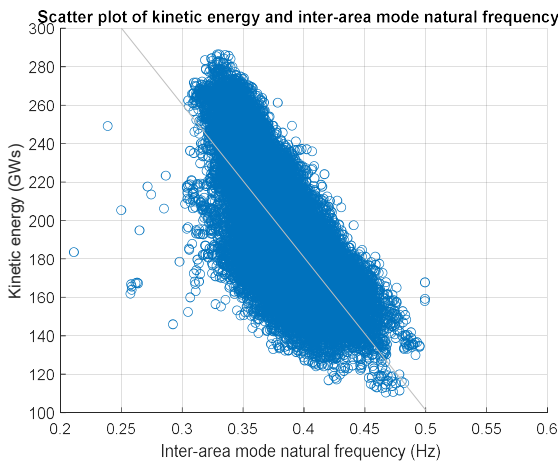


Fig. 6. Scatter plot of kinetic energy and natural frequency of the Finland-Sweden mode (the grey line is a linear fit between the two variables).

The correlation between the damping ratio of the Finland-Sweden mode and the kinetic energy was also analyzed. There was, in practice, no observable correlation between these two variables.

IV. CONCLUSIONS AND FUTURE WORK

This paper analyzed the characteristics of the main inter-area mode between Finland and Sweden during a 5-year period, based on measurements. The frequency and damping of the mode were identified from ambient measurements using a Bayesian modal identification method (mode meter). Furthermore, the paper analyzed the kinetic energy of the Nordic power system from the same 5-year period and investigated the relationship of the kinetic energy and the modal parameters of the inter-area mode.

The results of the analysis indicate:

- Kinetic energy of the Nordic system and the natural frequency of the Finland-Sweden inter-area mode have a clear relationship: When the kinetic energy is low, the natural frequency is high, and vice versa. The correlation coefficient between the two quantities was -73 % during the analysis period. The relationship seemed to hold during the whole 5-year analysis period.
- There was a decreasing trend in the kinetic energy during the 5-year period. This is likely due to synchronous generators being replaced by wind power. At the same time, an increasing trend in the inter-area mode frequency was observed. When the frequency of the inter-area mode changes significantly (which likely happens in the future), power system stabilizers (PSS) and other control systems, such as POD (Power Oscillation Damping) controls of SVCs, have to be re-tuned. As a practical step, it is important for transmission system operators to consider these changes in inter-area mode frequencies and update their requirements for PSS and other relevant control systems.

- The damping of the Finland-Sweden inter-area mode did not have a relationship with the inertia.

In the future, the kinetic energy (or inertia) of several different power systems is expected to decrease more. Therefore, it is highly important to monitor the inertia of the system and analyze how it effects other phenomena in the system. As kinetic energy seems to have a clear relationship between the frequency of the inter-area modes, this relationship can be utilized to estimate the kinetic energy of the system in real-time. However, due to inaccuracies there is likely a need to include also other variables in the estimation to improve the accuracy (using models with several variables). These variables could include, for example, frequency estimates of several inter-area modes, power flow measurements, load measurements, connection status of grid components, etc. Developing a model for online inertia estimation, based on inter-area mode frequencies and other (measured) variables, is proposed as a topic for future research.

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